

# Market Analysis of AI Image Generator Application

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## *Abstract*

This project explores the integration of generative artificial intelligence (AI) in digital marketing and content creation, focusing on applications for small businesses. The primary aim is to harness the capabilities of text-to-image generation technologies to streamline and enhance marketing and advertising efforts such as advertisements, social media content, and promotional graphics, at a much reduced costs.

A user-friendly platform, that allows users to input textual descriptions of desired visuals, which the AI then converts into high-quality images using state-of-the-art machine learning model like Stable-Diffusion. The project explores an idea of building a product and developing a possible business model out of such a product.

## **1. Problem Statement**

Small businesses often struggle to create high-quality, engaging marketing and advertising content due to limited resources, lack of design expertise, and budget constraints. Traditional content creation processes are time-consuming, expensive, and require specialised skills that small business owners may not possess or afford. This results in suboptimal marketing efforts, reduced audience engagement, and ultimately, hindered business growth. At this age of internet missing out on the opportunity to do promotions on social media platforms would be disadvantageous for the businesses. I, thus try to tackle this problem with Generative AI, specifically a text-to-image generation application, which offers a promising solution to these challenges by enabling the automated creation of visually appealing and customised marketing materials at a higher degree of control, customisability, much reduced cost and efforts.

## 2. Market/ Customer Needs Assessment

### 2.1 Market Analysis

The global digital marketing industry is projected to grow at a compound annual growth rate (CAGR) of 17.6% from 2021 to 2026. Small businesses constitute a significant portion of this market, with millions of SMEs globally seeking effective digital marketing solutions, which can help them compete with larger enterprises. There is an increasing demand for high-quality digital content, with visual content being a critical component, which can get up to 94% more views than content without them. The rise of social media platforms and visual-based marketing channels like Instagram and Pinterest highlights the need for visually appealing content.

### 2.2 Customer Segmentation

Typically, small business owners have limited time, resources and have tight marketing budgets and cannot afford expensive design software or hiring professional designers. They need affordable, easy-to-use tools to create marketing content without requiring extensive design skills. There is a need for intuitive tools that can produce high-quality visuals with minimal input.

Businesses such as local restaurants, boutiques, and service providers require regular updates to their marketing content to engage with their community and attract new customers. They need tools that allow for quick turnaround times and flexibility in content creation.

Maintaining a consistent brand image across all marketing channels is challenging without professional design help. Small businesses need tools that help them easily incorporate their brand elements (logos, colours, fonts) into their content. Automated and quick content generation is essential to allow them to focus on other critical aspects of their business.

## 3. Target Specification

### 3.1 Core Functionality and Design

**Text-to-Image Generation:** Converts user input text descriptions into high-quality images. Supports various content types (advertisements, social media posts, promotional graphics).

**Customisation:** Allows users to add logos, brand colours, fonts, and other brand elements. Provides editing tools for further image refinement.

**Intuitive Design:** Simple, user-friendly interface suitable for users with no design experience.

## 3.2 Performance Requirements

**Speed and Efficiency:** Text-to-image generation should be completed within a short period of time to ensure a smooth user experience. 99.9% uptime to ensure reliability for users relying on the platform for regular content creation.

**Scalability:** Must handle multiple users simultaneously without performance degradation. Scalable infrastructure to support growing user base and increased content generation demands.

**Quality:** Generated images must be of high quality suitable for various digital platforms. Consistency in visual quality across different types of content.

## 4. External Search

The product requires usage of deep learning models, which are really hard to write from scratch in order to suit our needs. Tweaking these models for our use is less daunting than coding it up from scratch. A pre-trained model can be used with the help of transfer learning methods. A lot of such pre-trained models are there which are open source and available in <https://huggingface.co/models>. One such model based on Stable Diffusion technique by StabilityAI can be used.

### 4.1. Benchmarking

In the market of text-to-image generation there are several already available models like DALL-E, DALL-E 2.0 from OpenAI. There is also Midjourney based on Stable Diffusion which are not free. My idea is to build a text-to-image generator product which, even though can be repurposed for any task, is well suited for small businesses and for integration to any software system through API. For the model itself, I have used “stable-diffusion-v1-4” from CompVis which is available in Huggingface. This particular model showed better stability in image generation compared to others in my system.

## 5. Constraints and Regulations

Several constraints must be kept in mind-

- The image generation depends on server performance, thus a high performance server is required.
- The model automatically filters out explicit content which is an advantage.
- Data protection and privacy regulations (Customers).

- Govt Regulations for small businesses.

## 6. Business Model

A business and monetisation model can be developed around such a product geared towards small businesses and even for individual users.

A subscription based model would be well suited for such a product. A tiered subscription plan e.g Basic, Pro, Enterprise to cater to different business sizes and needs and even individual creators, users. Competitive pricing to ensure affordability for small businesses. Offer a free tier with basic features and limited image generation to attract users. Encourage users to upgrade to paid plans for more features and higher usage limits.

A Basic plan with limited features and a cap on the number of images generated per month. Ideal for solo entrepreneurs and very small businesses.

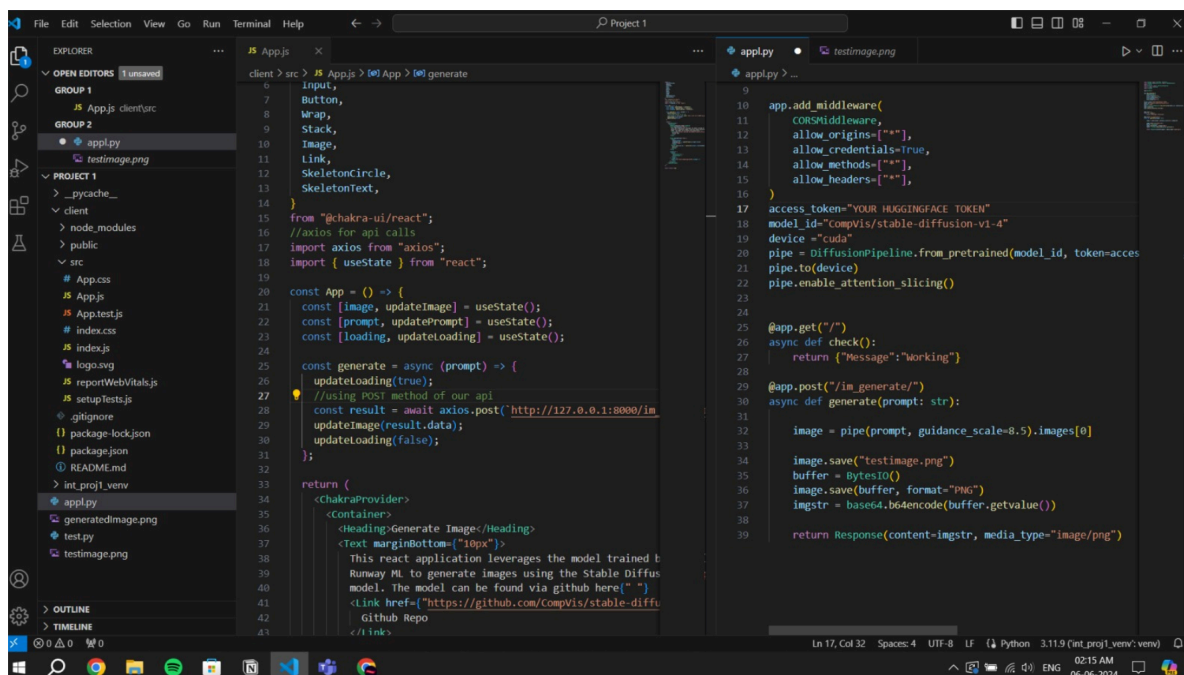
A Pro plan with unlimited image generation, advanced customisation options, and priority customer support. Suitable for growing small businesses and freelancers.

An Enterprise plan with custom solutions, dedicated account manager, team collaboration features, and API access. Designed for marketing agencies and larger small businesses.

In-platform advertising can be used Display non-intrusive ads within the platform, such as relevant marketing tools, services and also sponsorships.

## 7. Concept Generation and Development

In order to build such a product a usual and simple front-end interface with a high performance back-end with availability of API integration can be a viable option. Huggingface is a platform where, there are a number of pre-trained models available.

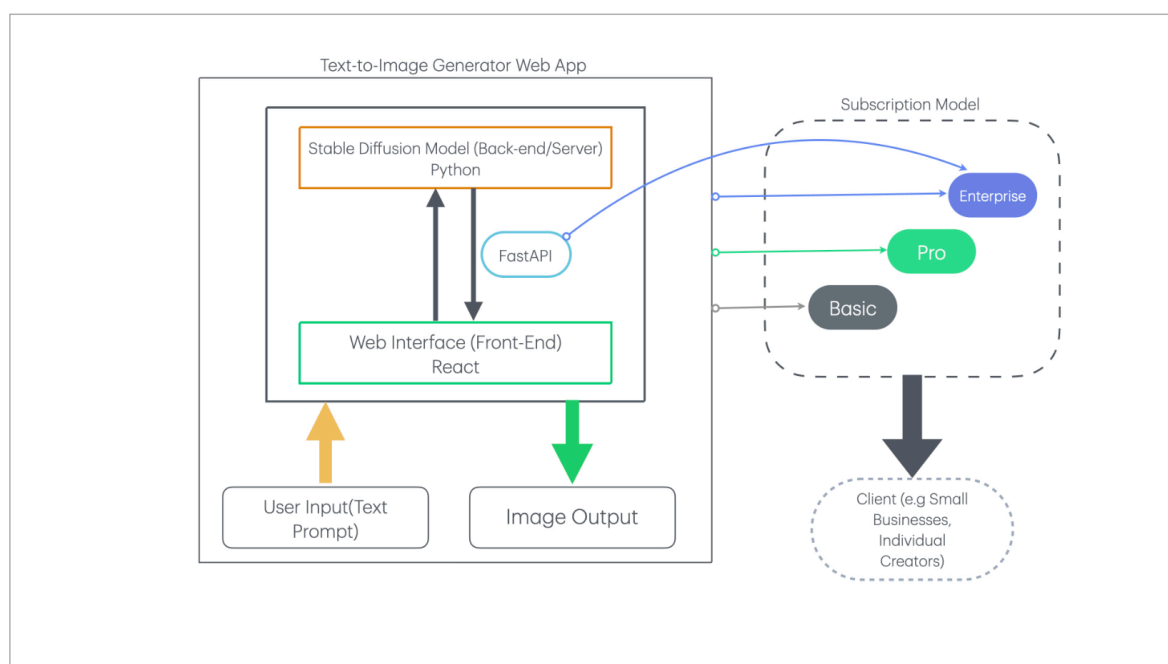


A portion of the Frontend and Python backend code

The Stable Diffusion model runs in a server which acts as a backend and a front-end or a software component can provide text prompt and obtain the generated image. The frontend is built using React framework in JavaScript and Backend using python and FastAPI according to the need.

## 8. Final Product Prototype

The product has the following components and functions to provide satisfactory results.



Schematic Diagram of the Final Product Prototype

**Front-End:** The front-end interface, which the client will interact with, is build using React Framework and JavaScript. It includes a basic Header and Input text field and a Generate button which triggers the Back-end through API. The received JSON response has the encoded image, which is decoded and displayed on the client side interface.

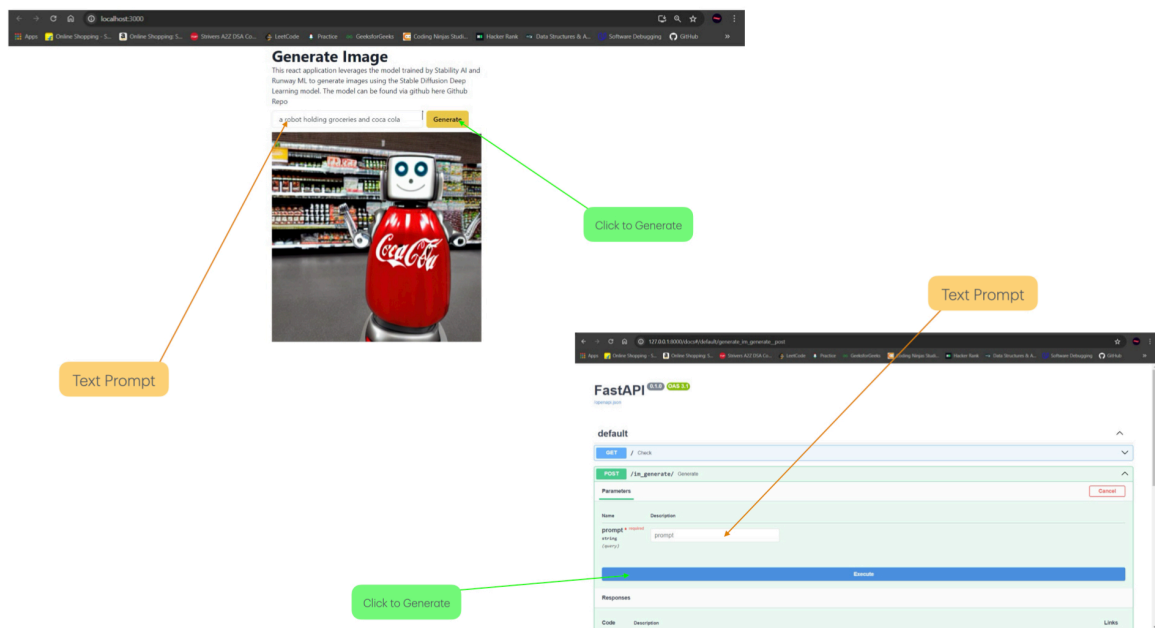
**Back-End:** The back-end is build in python language and FastAPI is used for building and generating the api interface. The Generate button in front-end triggers the server side python code which generates an image using the Stable Diffusion model and returns an image encoded in JSON. The Stable Diffusion uses PyTorch.



## 9. How Does it Work?

The client side will have an interface to input text and a generate button to start generation. To access for client side-

- The client needs to input the prompt.
- Click the generate button
- Generated Picture appears below

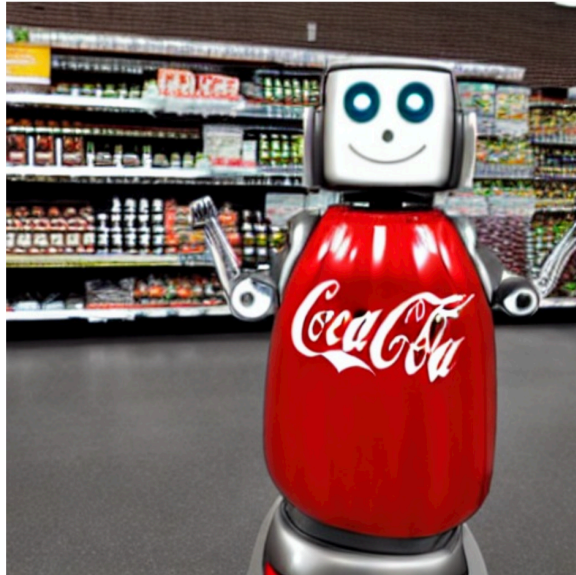


To access using api-

- At the given api link, user will get a FastAPI interface as above.
- User needs to click "Try Out" at the POST method and give the prompt.
- Then click the execute and the image will be generated at server side.

## 10. Implementation

Using the web app developed, I have been able to generate various images giving text prompt input. Some sample images are given below-



Prompt: a robot holding groceries and coca cola



Prompt: a cycle

The code implementation is uploaded in my GitHub.

The GitHub link- [https://github.com/iamabhijit01/feynnlab\\_project01\\_text2image](https://github.com/iamabhijit01/feynnlab_project01_text2image)

## 11. Financial Modelling and Equation

A possible business model has already been developed in the previous section where a subscriptions model has been chosen. Here I try to develop a Financial equation which basically shows how the Revenue growth that happens on the basis of sales.

### 11.1. Market Sales Analysis and Forecast

Due to very limited amount of sales data available for AI Image Generator applications a Time Series Forecasting or Market Segmentation analysis could not be done manually and had to be reliant on online sources.

Before developing a financial equation, it is important to understand market growth as it influences the sales of the product and in turn the produces revenue.

The AI Image generation market is expected to grow at a CAGR of 17.4% from 2023- 2030. The highest growth predicted in asian markets with a CAGR of 19.5% at the same time period. This in turn shows an exponential growth opportunity in sales of the product.

## 11.2. Financial Equation

The Basis of financial equation depends on a simple principle that the Generated revenue is directly proportional to the sales minus the expenses.

Considering Revenue to be -  $y$

Sales(a function of time itself)-  $x(t)$

The equation for revenue is-

$$y = cost * x(t) - expenses$$

**The expenses include:** the cost for the license of a Bare-bones Stable Diffusion Model, Cost for renting or maintaining a server to run the backend, also the front-end needs maintenance and periodic overhaul to keep the product new and relevant. Expenses for periodic analysis for expanding the product and develop new features. Also salaries for the employees must be included. A one-time cost for developing an algorithm from scratch for image generation and in house servers and GPUs can be planned for future at added cost. The Yearly Expenses in the initial stages can be around Rs. 2 lakh.

**For the cost:** We have various price ranges for the tiered subscription model. Initially, the costs could be-

Basic Tier- Rs. 89

Pro Tier- Rs. 249

Enterprise Tier- Rs. 799

Considering the market and subscription tiers the average cost may lie around 400 due a lot of the customers and businesses requiring Basic or Pro Tiers more and less of the Enterprise Tier. Therefore, the possible financial equation showing Yearly Revenue comes out to be- Yearly Revenue:

$$y(t) = 400 * x(t) - 200000$$

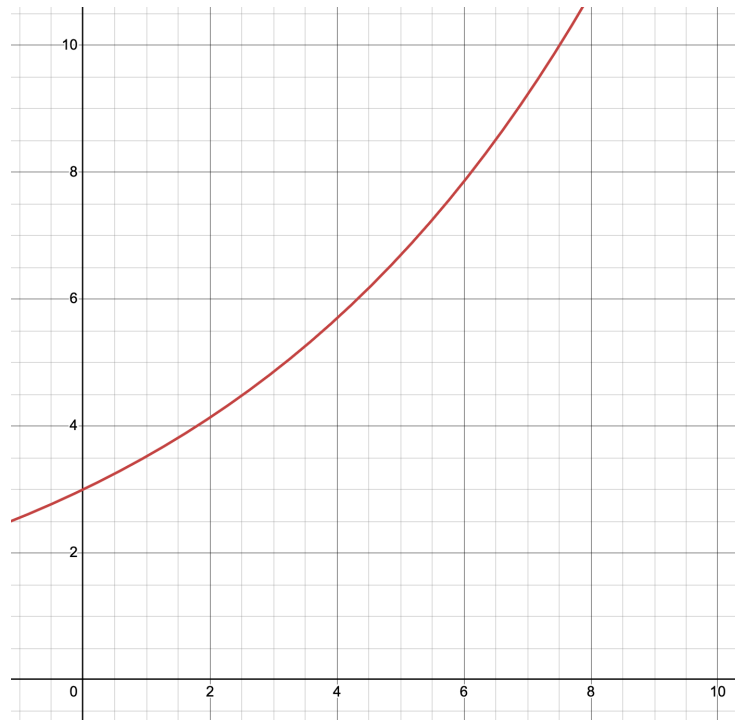
This is a linear revenue growth equation, but the sales value  $x(t)$  varies exponentially with time due to current market trend and forecast. Considering Compound growth of market and sales the growth can be shown via the equation-

$$x(t) = x_0(1 + rate)^t$$

At a current annual growth rate of 17.4% (CAGR) we have-

$$x(t) = x_0(1 + 0.174)^t$$

Therefore, the market grows 1.174 times each year from previous year. Current market size of AI image generator is about  $x_0=300,000$  US Dollars and about 900,000 US Dollars by 2030.



Market Size (in \*100,000 USD) vs Time(in Years)

## 12. Market Segmentation

The Market for such an AI image generator, even though at a very early stage, can be studied and segmented into groups which helps in targeting specific groups to focus the product to. Due to such products being at a very stage in the market, I could not find datasets to perform Segmentation analysis on my them and had to resort to different resources available online to identify possible segments and the segment in which the product falls.

**Pricing:** There are two categories - Free and Paid. Most of the available fully functional products are currently paid with the free versions being some of the open source algorithms for Text to Image Generator. This product can be categorised as a paid due to subscription model.

**End-User and Application:** The overall application areas include- Advertising, E-Commerce, Healthcare, Art, Fashion, Social Media. For this project the target was Small business and Individual Users. Therefor, Advertising, E-Commerce and Art are the use cases the product should aim for and build the specifications around those use cases.

**Geographic:** The largest markets for such a product are- North America, Europe, Asia Pacific, Middle East & Africa and Latin America. The Asian market shows the largest growth at a CAGR of 19.5% in the next 5 years. The product can, therefore, be targeted towards Asia especially in countries like India and China and also North America and Europe.

**Present Companies:** Another important point to note is the products that are available in the market- Dall-E2 by OpenAI, Midjourney, Adobe, Starry.ai are some of the biggest in the market. But a lot of other products for Google, Microsoft, Samsung and Apple are being developed that directly integrate Image Generative features into their respective operating systems as system softwares.

## 13. Conclusion

The AI-Marketer project revolutionises digital marketing for small businesses by using generative AI for text-to-image creation. This user-friendly platform addresses the challenges of limited resources, design expertise, and time, enabling small businesses to produce high-quality, customised marketing content easily and affordably.

With a robust business model featuring subscription plans, pay-per-use options, and premium features, AI-Marketer ensures sustainable revenue and growth. By democratising access to professional-grade marketing tools, AI-Marketer empowers small businesses to enhance their digital marketing efforts, engage audiences effectively, and drive business success.

With the ongoing expansion of Generative AI and related products, such an early market analysis, product prototype development and a financial model development are invaluable informations as it gives better understanding of the products in the growing phase of the market for Generative AI products which might give a company a great advantage as the market seems to grow to a great extent in the upcoming several years according to the market analysis.

## 14. References and Resources

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